

Systems that refuse to invent.

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Every system I use will guess, then deliver the guess in the voice it uses for facts. That isn't a bug – **an honest “I don't know” is harder to build than a confident answer**, and nobody is rewarded for building it.

So that is what I build. Five projects, one idea, arrived at five times before I noticed it was a thesis.

32 repositories · 829 contributions in twelve months · 29 patches merged into 35 open-source projects, approved by 10 maintainers.

01	02	03	04	05
Recall	ATTEST	Support Triage	Minutes	PDFChat
A memory engine for unfinished work	Money that moves only on proof	An agent that cannot be talked into it	A meeting workspace, rebuilt	Cites its pages, or refuses

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Recall

A memory engine for unfinished work. One keystroke brings back the tabs, files and chats from your last session.

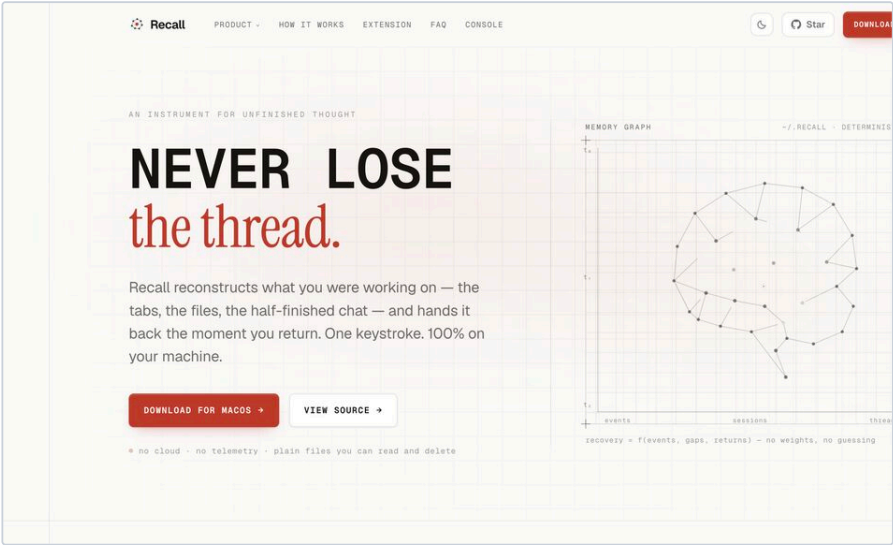
Someone interrupts you at 3pm. You sit back down and everything is still there – the file, the tab, the branch – and none of it helps. **You didn't lose files. You lost continuity.**

Spotlight and grep assume you remember a string. A cognitive restart never asks that. It asks *what was I doing* – and nothing on the machine answers that.

Press **Ctrl + Space**, type half a thought, and the moment, the session and the topic all come back at once. An empty query offers what to resume, **capped at four items** – showing you everything relevant is the same as showing you nothing.

TECHNICAL ASPECTS

Seven layers over a local event log, fed by a Chrome extension and a file watcher. **No model in the production path** – the same question can never return two different pasts, enforced by a 35-section smoke test. No cloud, no account, no telemetry: everything it knows sits in `~/recall/` as plain JSON, readable with `cat`. Ships as a 26 MB menu-bar app that reports degraded features instead of hiding them.



YEAR	2026
ROLE	Sole author – engine, launcher, extension, daemon, docs
BUILT WITH	Python · PyQt6 · FastAPI · ChromaDB · Tauri
LINK	web-seven-puce-99.vercel.app github.com/kunalKumar-13/Recall-me

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ATTEST

A financial control layer. It verifies what an AI proposes before anything is allowed to move money.

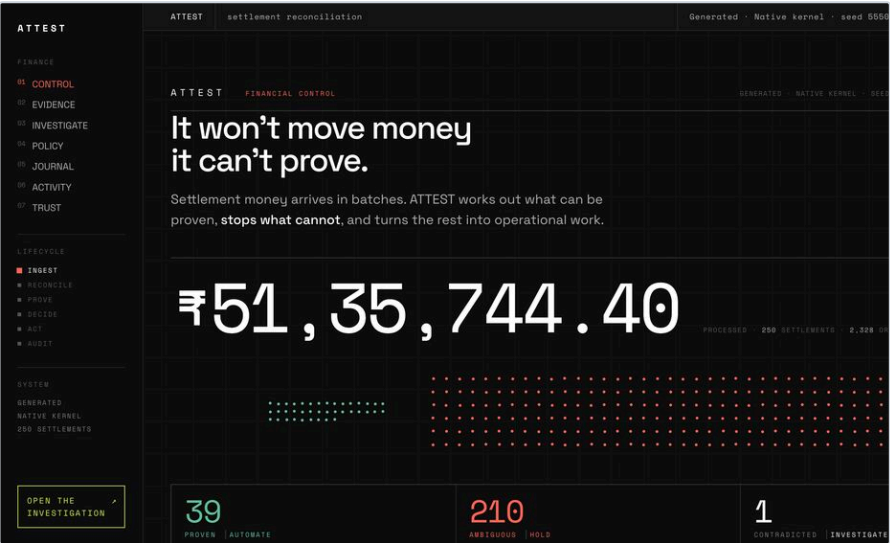
A payment gateway sends one lump sum covering many orders. Somebody has to work out what it was made of – and **more than one set of orders can add up to the same credit exactly**.

Those sets discharge *different customers*. Guessing marks the wrong customer as paid while the books still balance, so no alarm fires and nobody finds out until it's expensive.

So ATTEST refuses. Across 500 settlements it had never seen, **82.4% were correctly refused and ₹0.00 moved against the wrong account**. The refusal is the product, not a failure rate.

TECHNICAL ASPECTS

Four stages, and the separation is the design. An **advisor** proposes which records to look at, and is **never given an amount**. A **verifier** sharing no code with it re-derives every explanation independently. A **policy** automates only where measured error costs less than a human check. Integer paise end to end – no float ever touches money. The published benchmark names a competitor that beats mine on precision, because a comparison curated in your own favour proves nothing.



YEAR	2026
ROLE	Sole author – architecture, solver, evaluation
BUILT WITH	Python · Rust solver kernel
LINK	github.com/kunalKumar-13/attest

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Support Triage

A support agent where code owns the control flow and the model is a constrained analyst – not the decision-maker.

A support agent that can be talked into a destructive action is worse than no agent. The usual design hands the model the wheel and hopes the prompt holds.

This one doesn't. **The LLM is invoked exactly once per ticket, inside an unforgeable sentinel block**, and returns a typed JSON object that a policy validator adjudicates. It cannot decide escalation, cannot execute tools, and never sees raw PII.

Its worst possible behaviour is *escalate to a human*. **132 of 132 evaluation tests pass**, and two runs on the same input produce a byte-identical file.

TECHNICAL ASPECTS

Untrusted ticket text never leaves its frame; retrieved chunks are grounding, never instructions. Defends prompt injection across seven languages plus zero-width, homoglyph and base64 obfuscation. When retrieved chunks contradict each other, **confidence drops and the validator escalates** rather than picking a side. Every tool call is schema-validated, risk-gated and idempotency-checked.

```
$ triage --ticket 4417

  ingest    ticket text sealed in <<<TICKET>>>
  retrieve  4 chunks · 2 disagree numerically
  analyse   LLM called once, inside sentinel block
            → typed JSON, schema validated
  policy    consensus broken → confidence 0.41
            escalation cap 0.60 · not eligible

  DECISION  ESCALATE TO HUMAN
  reason    retrieval_conflict, low_confidence

$ sha256sum run_a.csv run_b.csv
2f9c...a71b  run_a.csv
2f9c...a71b  run_b.csv
```

YEAR	2026
ROLE	Sole author – pipeline, policy validator, threat model
BUILT WITH	Python · LlamaIndex · BM25 · TF-IDF
LINK	github.com/kunalKumar-13/deterministic-support-triage

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Minutes

A meeting workspace – interactive transcripts, AI notes and workspace-wide search – rebuilt from the product outward.

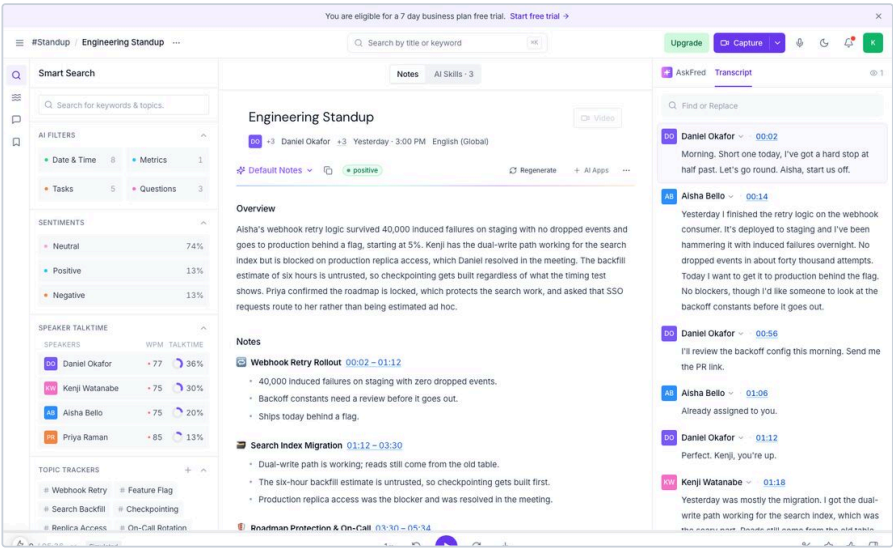
“Clone this product” is the easiest brief in the world to do badly. You rebuild the hero section and ship something that photographs well and collapses when anyone clicks.

But a meeting product isn't its landing page. It's what happens on the fourth day, when you half-remember someone committing to something about pricing and you need to find it. **Everything valuable happens after the meeting ends.**

So I set a bar I couldn't fake: everything except the speech-to-text actually works – and the README says so on the first screen, before a reviewer can discover it.

TECHNICAL ASPECTS

Next.js 15 over FastAPI, with **SQLite FTS5** as a real full-text index rather than a **LIKE** query in a costume. Auth is enforced at the edge, so a signed-out visitor never receives app UI. First boot seeds seven complete meetings – **git clone** to a populated product in two commands. The decision that mattered: I stopped matching screens and **extracted the token system instead**, after which twenty-plus screens came out coherent untouched.



YEAR	2026
ROLE	Sole author – frontend, backend, schema, search, summariser. Built to an assignment brief.
BUILT WITH	Next.js 15 · React 19 · TypeScript · FastAPI · SQLite FTS5
LINK	github.com/kunalKumar-13/minutes

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PDFChat

Ask a document anything. Every answer cites its pages – or says the document doesn't support the question.

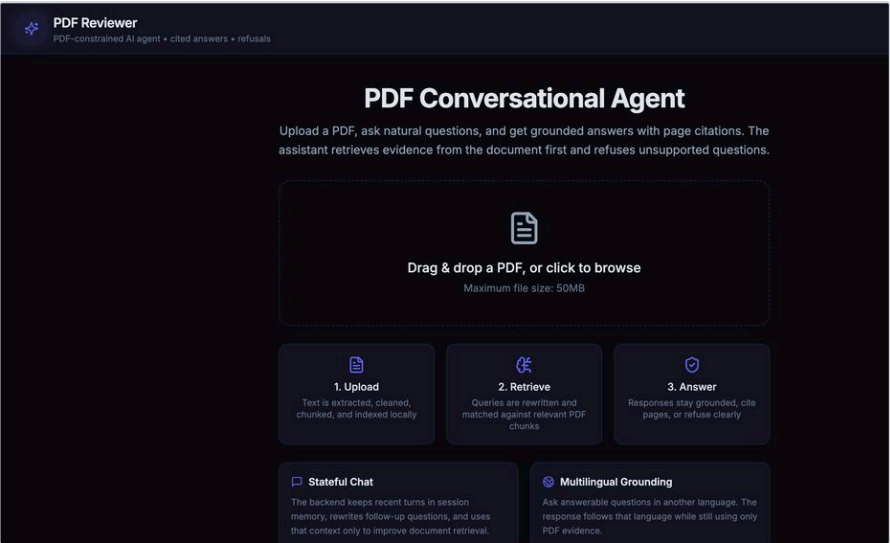
A confident wrong answer costs more than no answer. Retrieval systems get messy, and the standard failure is a model that fills the gap fluently and invisibly.

PDFChat won't. Every response either **cites the pages it came from** or refuses clearly. Multi-turn chat keeps session context for retrieval only – the document stays the sole evidence.

It answers in your language while still treating the source as the only truth – the case where grounding usually breaks quietly.

TECHNICAL ASPECTS

Extraction, chunking and indexing all run locally; queries are rewritten against the relevant chunks before generation. The refusal path is a first-class branch, not an error handler – if retrieval doesn't support an answer, that *is* the answer. Handles documents up to 50 MB, deployed and publicly usable.



YEAR	2026
ROLE	Sole author – retrieval, citation and refusal path
BUILT WITH	Python · PyMuPDF · TF-IDF retrieval · FastAPI
LINK	pdf-reviewer-ten.vercel.app github.com/kunalKumar-13/PDF-Reviewer

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